

Classify prime, square and cube numbers

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Classify 97, 144 and 125. Copy or complete the first step.

2. Is 91 prime? Explain.

3. Write the square numbers between 50 and 150.

4. Miro says: Miro says 1 is prime because it has no factor pairs apart from itself. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Classify 97, 144 and 125. Copy or complete the first step.

Answer 97 is prime; 144 is square; 125 is cube.

2. Is 91 prime? Explain.

Answer No, because $91 = 7 \times 13$.

3. Write the square numbers between 50 and 150.

Answer 64, 81, 100, 121 and 144.

4. Miro says: Miro says 1 is prime because it has no factor pairs apart from itself. Is this correct? Explain.

Answer A prime number has exactly two factors. The number 1 has only one factor.

Classify prime, square and cube numbers

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Classify 97, 144 and 125.

6. Check this result using an inverse, estimate or known fact: Write the square numbers between 50 and 150.

2. Is 91 prime? Explain.

7. Find a number below 200 that is both a square and a cube.

3. Write the square numbers between 50 and 150.

8. Identify and correct this misconception: Miro says 1 is prime because it has no factor pairs apart from itself.

4. Solve this independently and show every stage: Classify 97, 144 and 125.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Is 91 prime? Explain.

10. Write and solve a similar question to: Write the square numbers between 50 and 150.

Teacher answers and checking prompts

1. Classify 97, 144 and 125.
Answer 97 is prime; 144 is square; 125 is cube.
2. Is 91 prime? Explain.
Answer No, because $91 = 7 \times 13$.
3. Write the square numbers between 50 and 150.
Answer 64, 81, 100, 121 and 144.
4. Solve this independently and show every stage: Classify 97, 144 and 125.
Answer 97 is prime; 144 is square; 125 is cube.
5. Use a second method or representation for: Is 91 prime? Explain.
Answer Expected result: No, because $91 = 7 \times 13$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Write the square numbers between 50 and 150.
Answer Expected result: 64, 81, 100, 121 and 144. A valid check must be shown.
7. Find a number below 200 that is both a square and a cube.
Answer 64.
8. Identify and correct this misconception: Miro says 1 is prime because it has no factor pairs apart from itself.
Answer A prime number has exactly two factors. The number 1 has only one factor.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Write the square numbers between 50 and 150.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Classify prime, square and cube numbers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find a number below 200 that is both a square and a cube.

6. Create a new example that tests the same idea as: Write the square numbers between 50 and 150.

2. Prove the answer to this question, rather than only calculating it: Classify 97, 144 and 125.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Is 91 prime? Explain.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 64, 81, 100, 121 and 144.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro says 1 is prime because it has no factor pairs apart from itself.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find a number below 200 that is both a square and a cube.
Answer 64.
2. Prove the answer to this question, rather than only calculating it: Classify 97, 144 and 125.
Answer Expected result: 97 is prime; 144 is square; 125 is cube. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Is 91 prime? Explain.
Answer Expected result: No, because $91 = 7 \times 13$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 64, 81, 100, 121 and 144.
Answer One valid reconstruction is: Write the square numbers between 50 and 150. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro says 1 is prime because it has no factor pairs apart from itself.
Answer A prime number has exactly two factors. The number 1 has only one factor.
6. Create a new example that tests the same idea as: Write the square numbers between 50 and 150.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.